

Grade 12 Mathematics: Differential Calculus & Derivatives

1. Concept of Limits and Derivatives

Differential calculus studies how quantities change. The derivative of a function $f(x)$ represents its instantaneous rate of change.

Definition of Derivative

Using the first principle of limits:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

2. Standard Derivative Formulas

1. Power Rule:

$$\frac{d}{dx}(x^n) = n x^{n-1}$$

2. Exponential & Logarithmic Functions:

$$\frac{d}{dx}(e^x) = e^x, \quad \frac{d}{dx}(\ln x) = \frac{1}{x}$$

3. Trigonometric Functions:

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x, \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

3. Rules of Differentiation

Product Rule

$$\frac{d}{dx}[u(x) \cdot v(x)] = u \frac{dv}{dx} + v \frac{du}{dx}$$

Quotient Rule

$$\frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Chain Rule (Function of a Function)

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

4. Solved Example Problem

Problem: Find the derivative of $y = (3x^2 + 5)^4$ with respect to x .

Solution:

Let $u = 3x^2 + 5 \implies y = u^4$.